IT3120- Lecture 07

Legal and Ethical Issues in IT Business



Learning Outcomes

At the end of this lecture, students will be able to:

- Identify legal and ethical issues in IT scenarios
- Understand software licensing requirements
- Recognize data privacy principles in practice.

Legal Issues in IT Businesses

Legal issues in Information Technology (IT) refer to the legal challenges and considerations that arise in the development, implementation, and use of computer technologies.

- Intellectual Property
- Data Privacy
- Software Licensing
- Cybersecurity Laws
- Electronic Contracts and E-Signatures
- Regulatory Compliance

Intellectual Property Rights in IT

- **Patents** – Protect new inventions, technologies, and processes.
- **Copyrights** – Protect software code, documentation, and other original works.
- **Trademarks** – Protect brand names, logos, and distinctive identities.
- **Software Licensing** – Defines how software can be used, modified, and distributed.
- **Open-Source Software** – Requires compliance with specific licensing terms and conditions.
- **Proprietary Technology** – Protects confidential and commercially valuable technologies.

Data Privacy

- **Protect Personal Data** – Keep personal information safe from unauthorized access or misuse.
- **Respect Privacy** – Respect individuals' rights over their personal information.
- **Responsible Data Handling** – Collect, store, use, and share data appropriately.
- **Follow Data Protection Laws** – Organizations must comply with relevant privacy and data protection regulations.
- **Prevent Data Breaches** – Take appropriate measures to protect data from loss, theft, or unauthorized access.

Software Licensing

Software licensing defines how software can be used, copied, modified, and distributed.

Software License Specify:

- Who can use the software
- How the software can be used
- Whether it can be copied or shared
- Whether it can be modified
- How it can be distributed

Software Licensing - Types

Proprietary – owned by a company; restrictions on use, modification, sharing.

Open-Source – source code available; must comply with license terms (e.g., GPL, MIT, Apache).

Freeware – free to use, but source code not always open.

Shareware – trial version; payment required for full features.

Cybersecurity Laws

Cybersecurity laws are laws and regulations designed to protect computer systems, networks, and data from unauthorized access, misuse, and cyberattacks.

These regulations define how organizations must protect data from breaches, unauthorized access, and misuse.

Example: Uber Breach (2016)

Electronic Contracts and E-Signatures

Electronic contracts and electronic signatures are important for conducting business and transactions online.

- **E-Contracts** – Agreements created, accepted, and stored electronically.
- **E-Signatures** – Electronic methods used to indicate agreement or approval.
- **Legal Validity** – E-contracts and e-signatures can have legal recognition under applicable laws.
- **Security and Authenticity** – Measures should be taken to verify identities and protect electronic transactions.

Regulatory Compliance

Regulatory compliance means following the laws, rules, and standards that apply to an organization or industry.

These rules help protect customers, employees, and business information. Failure to follow these rules can result in fines, legal action, or damage to the organization's reputation.

Examples:

- Finance - PCI-DSS
- Healthcare - HIPAA
- Telecom - TRCSL

Ethical issues related to IT professionals

IT professionals play an important role in developing and managing technologies that affect individuals, organizations, and society. With this responsibility comes the need to make ethical decisions and use technology responsibly.

- **Personal Privacy**
- **Access Right**
- **Harmful Actions**
- **Trade Secrets**
- **AI & Automation**

Personal Privacy

Privacy and confidentiality are important ethical issues in Information Technology because IT systems handle large amounts of personal and sensitive information.

- **Protect Personal Information** – Prevent unauthorized access to users' data.
- **Respect User Privacy** – Use personal information responsibly and only for appropriate purposes.
- **Maintain Data Integrity** – Ensure that information remains accurate and is not improperly changed.
- **Prevent Unauthorized Disclosure** – Do not share confidential information with unauthorized individuals.
- **Use Data Responsibly** – IT professionals should respect individuals' privacy when collecting, storing, and using data.

Access Right

Access rights determine who is allowed to access systems, applications, and information, and what they are allowed to do with them.

- **Authorized Access** – Only approved users may access systems and data.
- **User Permissions** – Users should only have access to the information they need.
- **Unauthorized Access** – Accessing someone else's account or data without permission is unethical and may be illegal.
- **Access Control** – passwords, MFA, role-based permissions.
- **Monitoring and Detection** – detect suspicious activity; but balance with privacy.

Harmful Actions

Harmful actions in IT refer to using technology in ways that cause damage, loss, or negative consequences to individuals, organizations, or society.

- **Spreading Malware** – viruses and other malicious software.
- **Destroying Data** – Intentionally deleting or damaging important files.
- **Altering Systems** – Making unauthorized changes to software or computer systems.
- **Damaging or Misusing Resources** – Using IT resources in ways that cause financial or operational loss.
- **Causing Disruption** – Actions that prevent systems or services from functioning properly.

Trade Secrets

Trade secrets are confidential business information that provides an organization with a competitive advantage and is not publicly known.

- **Source Code** – Confidential software code developed by a company.
- **Algorithms and Methods** – Unique technical processes or solutions.
- **Business Information** – Confidential strategies, plans, or customer information.
- **Research and Development** – Unreleased technologies or product designs.

AI & Automation

AI can provide many benefits, but it also creates ethical challenges:

- **Bias and Fairness** – AI systems produce unfair or discriminatory results if trained on biased data.
- **Human Responsibility** – critical decisions must involve human judgment.
- **Impact on Employment** – AI and automation may change or replace some jobs.
- **Transparency** – Users should understand how and why AI systems make important decisions.
- **Safety and Security** – protect AI from misuse

Real World Examples related to Legal and Ethical Issues

Cambridge Analytica / Facebook data controversy

Personal Facebook data were obtained and used for political profiling without users' informed understanding of how their data would be used. The case raised major questions about privacy, consent, data protection, and responsible use of personal information.

- Legal: Violation of data protection rules (GDPR fines).
- Ethical: Lack of transparency, misuse of trust.

Reference:

https://greatermanchester.ac.uk/blogs/the-cambridge-analytica-scandal-and-what-it-teaches-us

Uber's 2016 data breach

Uber experienced a major data breach in which information relating to millions of users and drivers was accessed. The incident raised issues around cybersecurity, disclosure of breaches, responsibility, and protection of personal information.

- **Legal:** Data protection and breach notification requirements.
- **Ethical:** Responsibility to protect users' information and be transparent about security incidents.

Reference:

https://www.huntress.com/threat-library/data-breach/uber-data-breach

Amazon's AI recruitment system

Amazon developed an AI-based recruitment tool that reportedly showed bias against women because it learned from historical hiring data.

- **Legal**: Potential discrimination concerns.
- **Ethical**: AI should provide fair and non-discriminatory outcomes.

Reference:

https://www.reuters.com/article/world/insight-amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK0AG/

Pratibha Syntex – Use of Pirated Software

An Indian textile manufacturer, Pratibha Syntex Ltd., was accused of using unlicensed software from companies including Microsoft and Adobe. The company settled the case for $100,000 and was required to stop using unlicensed software and conduct software audits.

- **Legal issue**: Copyright and software licensing
- **Ethical issue**: Using software without properly respecting the developers' intellectual property.

Reference:

https://in.thedollarbusiness.com/news/us-court-imposes-100000-penalty-on-indian-textile-firm/37641

Best practices for IT Professionals

IT best practices are proven methods, guidelines, and processes that help IT professionals perform their work effectively, securely, and efficiently.

- Strong Passwords & Authentication
- Regular Data Backups
- Software Updates & Security Patches
- Cybersecurity Awareness
- Access Control
- Data Protection & Privacy
- Proper Documentation
- Compliance with Laws & Policies

Computer Society of Sri Lanka (CSSL) Code of Ethics

Core Values of the CSSL Code of Ethics

1. Professionalism

- Maintain high professional standards.
- Behave with dignity and respect.
- Protect the reputation of the ICT profession.

2. Honesty

- Be truthful in professional work.
- Avoid plagiarism and deception.
- Give proper credit to others.

3. Competence

- Accept only work you are qualified to perform.
- Continuously improve technical knowledge.
- Deliver quality software and services.

Computer Society of Sri Lanka (CSSL) Code of Ethics

4. Professional Development

- Keep up with emerging technologies.
- Engage in lifelong learning.
- Share knowledge with colleagues and society.

5. Adding Value to Sri Lanka

- Use ICT to improve society.
- Support national digital transformation.
- Develop innovative solutions for national development.

6. Responsibility and Integrity

- Protect confidential information.
- Avoid conflicts of interest.
- Act in the best interests of clients, employers, and society.

Further Reading

- CSSL Code of Ethics
- Electronic Transactions Act No. 19 of 2006 (Sri Lanka)
- Computer Crime Act No. 24 of 2007 (Sri Lanka)
- GDPR Regulation (EU) 2016/679
- SLCERT – Sri Lanka Computer Emergency Readiness Team



In Class Activity

Question 01

Which type of intellectual property protects software code?

a) Patent
b) Copyright
c) Trademark
d) Trade Secret

Question 02

What is the primary purpose of data privacy laws?

a) To make money for the government

b) To protect individuals' personal information

c) To limit internet usage

d) To protect companies' data

Question 03

State whether the following statements are True or False:

a. Open-source software can never be used for commercial purposes.

b. Data breaches must always be reported to authorities in Sri Lanka.

c. The CSSL Code of Ethics is legally binding for all IT professionals in Sri Lanka.

d. E-signatures are legally valid in Sri Lanka.

e. It is ethical to use pirated software if you're a small startup with no budget.

THANK YOU